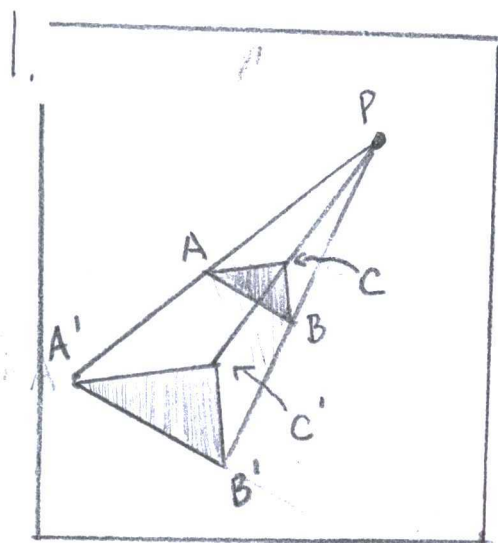


Euclidean and NonEuclidean Geometries:

Advanced Projects in the Projective Plane:



"Desarguesian
Plane"

Generally, Desargues planes project correct perspective. A Flashlight illuminates a circle and larger circles after further distance. Although, Desargues is not correct, in all circumstances, such as refraction. Sometimes light transmits into a material with large refraction. The lines project to a new shape.

Dual Desargues Statement, "

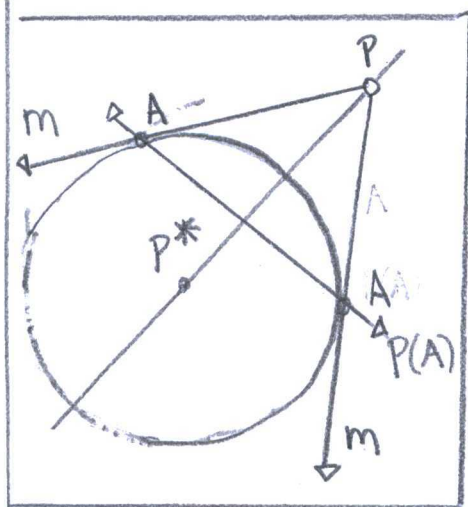
Two sets of vertices of two triangles correspond in such

a way that the lines joining corresponding vertices are concurrent, then the intersections of corresponding sides are collinear."

Converse to Dual-Desargues:

"Intersections of sides are collinear implies one or more vertices of two triangles with concurrent lines joining vertices."

2



"Polarity of P"

A homogenous quadratic equation in three variables forms into an equation $x^2 + y^2 - z^2 = 0$. The x^2 and y^2 represent a circle, while the z a radius to the circle. When the z -axis increases

the radius is 0. However, at $z=0$, the radius is zero and also vertex about a cone. The homogeneous equation for circles could model a cone.

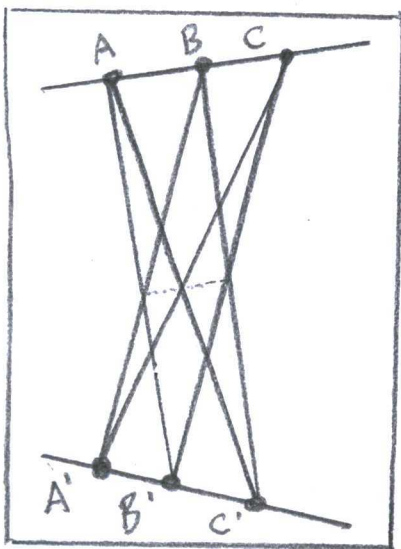
Extensions to conics include

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz = 0.$$

The large extension scales variables by A, B, C, D, E , and F . In summary,

"Polars of $\mathcal{P}$ " describes circles and cones.

3.



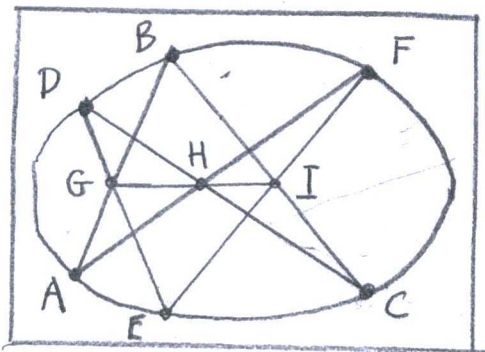
"Pappus
Theorem"

(from exercise) "If A, B , and C are three distinct points on

one line, and if $A', B',$ and C' are three other distinct points on a second line, then the intersections of lines AC' and CA' , AB' and BA' , and BC' and CB' are collinear."

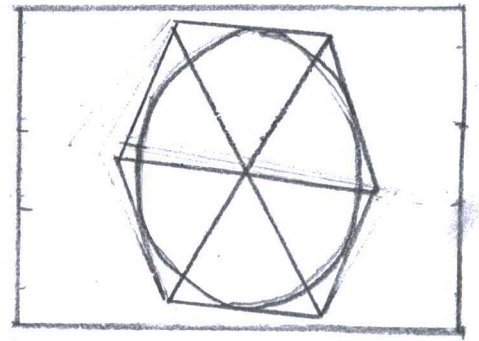
Pappus' statement implies Desargues in larger coordinate systems. A collinear line in Pappus' statement is possible in elliptic, hyperbolic, or Euclidean geometry. The line by Pappus becomes an edge to our shape in 3D-coordinates.

4.



"Pascal's mystic hexagram theorem"

"Brianchon's Theorem"



Pascal's Hexagram Theorem:

"If A, B, C, D, E , and F form line segments around a conic, then three points lie on a straight line."

Brianchon's Theorem:

"A circle circumscribed by a hexagon meets the circle at one point."

5. Definitions:

Skew field - a set containing values divisible by another element in the set and equating to an element in the set.

Ternary ring - a set mapping coordinates,
such as Desargues'
projection

6. The author presented "Duality" by a
Second set. Desargue's dual is two
projections or two "perspectives"
to an art student. But a
second set is proper to math.
A dual to Pappus theorem is
a set of points or set of lines.
Fano's dual is a set with
seven points or lines, and also
pair of points for each line.

7. Mathematicians look for answers to
theory by exhaustive computation.
Clement Lam resolved no solution
for "Projective Planes of order 10."